Amazon Lex and Conversational AI – Introduction

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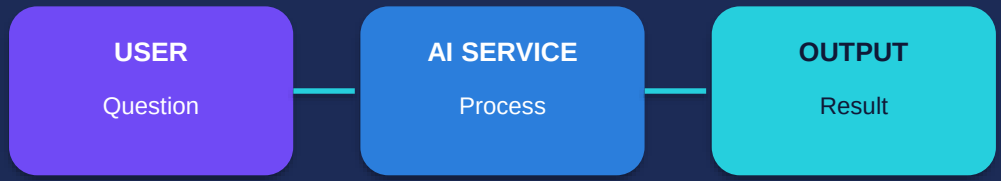
Amazon Lex and Conversational AI – Introduction

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Conversational interfaces**
- Text and speech experiences
- Intent recognition
- Dialog management
- Fulfillment and integrations

VISUAL CONCEPT | Conversational AI application architecture.



INPUT → PROCESSING → USEFUL RESULT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

This presentation focuses on Amazon Lex and how conversational AI can support chatbots and voice-enabled applications. It follows the course topics and highlights the key concepts used to design and test a bot.

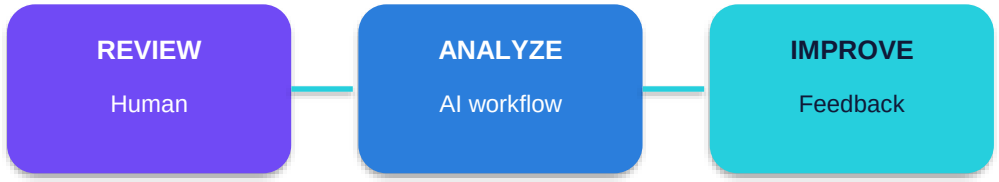
Understanding Intents and Utterances

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Intent represents user goal**
- Sample utterances
- Recognize varied phrasing
- Map requests to actions
- Test ambiguous inputs

VISUAL CONCEPT | User utterances mapped to intents.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

An intent represents what a user wants to accomplish. Sample utterances provide examples of how users may express that goal, and testing helps reveal gaps in recognition.

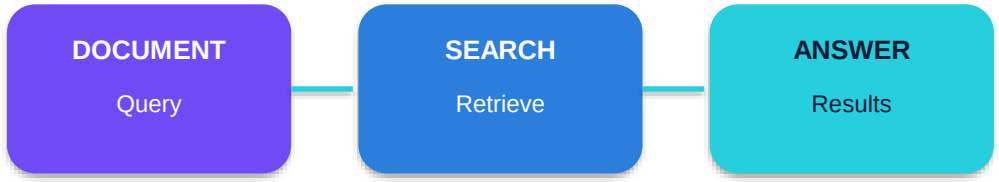
Slots, Prompts and Required Information

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Slots capture details**
- Slot types
- Required versus optional fields
- Prompts request missing data
- Validate collected values

VISUAL CONCEPT | Dialog collecting name, date, or other required fields.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Slots capture information needed to complete an intent. Prompts can request missing details, while validation helps ensure that collected values meet the application's requirements.

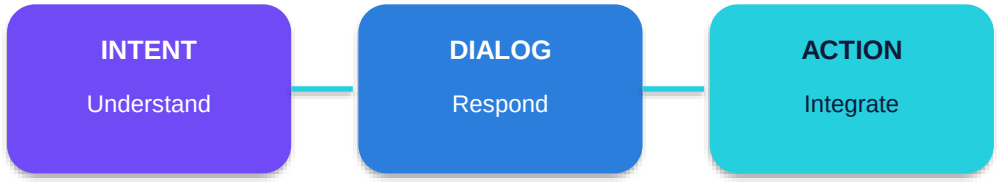
Dialog Management and Fulfillment

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Dialog steps**
- Confirmation
- Fulfillment logic
- Error handling
- User-friendly responses

VISUAL CONCEPT | Intent → collect slots → confirm → fulfill.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Dialog management organizes the conversation, while fulfillment performs the task after the required information is collected. Good design includes confirmation, error handling, and a clear next step for the user.

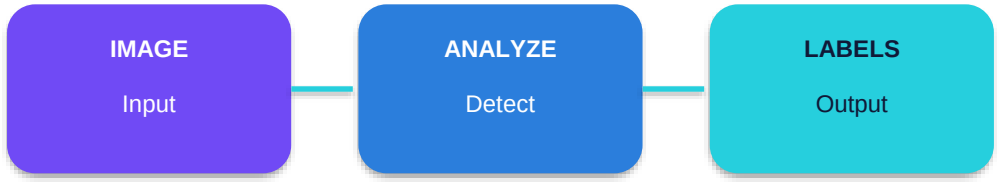
Amazon Lex – Hands-On Bot Configuration

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Create or open a bot**
- Configure language settings
- Add intents and utterances
- Configure slots
- Build and test

VISUAL CONCEPT | Bot configuration checklist and test loop.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A hands-on exercise helps learners understand the sequence of configuring and testing a Lex bot. Console labels and workflows may change, so follow the current AWS interface and course instructions.

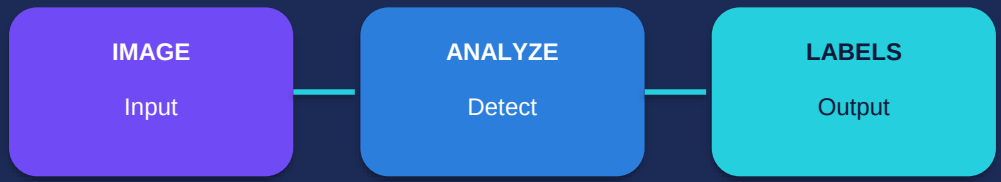
Testing Conversational Paths

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Happy-path testing**
- Missing information
- Unexpected responses
- Ambiguous utterances
- Fallback behavior

VISUAL CONCEPT | Conversation test cases with pass/fail outcomes.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Testing should cover both expected and unexpected user inputs. Reviewing fallback behavior and missing information helps make the bot more useful and less confusing.

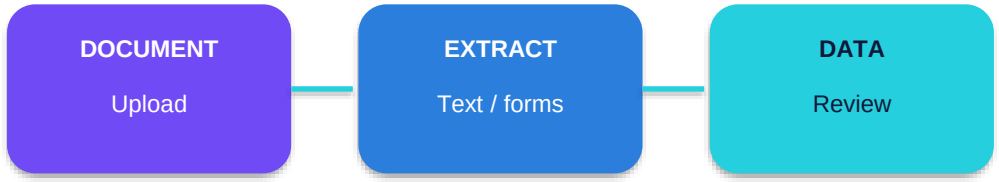
Amazon Rekognition – Overview

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Image and video analysis**
- Labels and objects
- Supported detection features
- Structured API responses
- Responsible use

VISUAL CONCEPT | Media input → Rekognition operation → structured output.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Amazon Rekognition analyzes supported image and video inputs and returns structured results. The chosen operation determines the type of analysis, and output should be validated for the intended use.

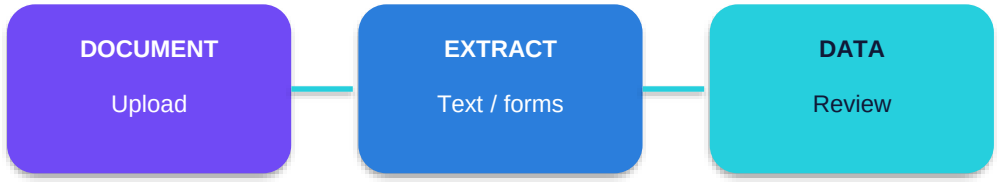
Amazon Rekognition – Hands-On Workflow

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Choose supported operation**
- Provide sample image or video
- Inspect response fields
- Review confidence values
- Validate results

VISUAL CONCEPT | Sample image analysis with output labels.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A practical exercise shows how to submit media to a supported operation and inspect the returned results. Learners should distinguish model output from verified facts and evaluate performance on representative samples.

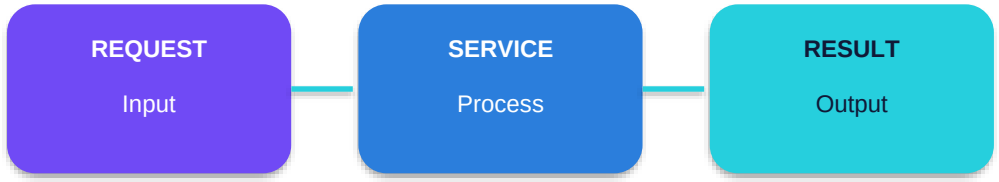
Using Rekognition Results in Applications

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Parse API response**
- Display useful results
- Handle errors and limits
- Protect sensitive media
- Monitor quality

VISUAL CONCEPT | Application consuming and displaying analysis results.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Applications should handle API responses carefully, present results with appropriate context, and account for errors and service limits. Sensitive media should be protected according to policy and applicable law.

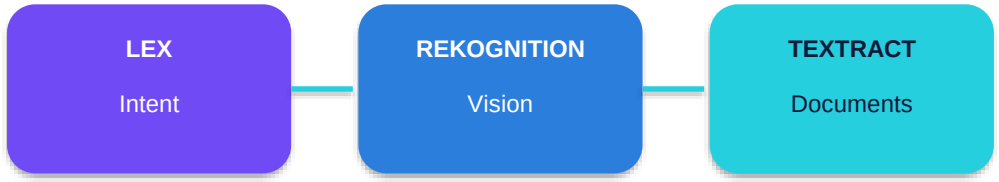
Comparing Lex and Rekognition

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Lex: understand user requests**
- Rekognition: analyze media
- Different input types
- Different output structures
- Combine only when useful

VISUAL CONCEPT | Lex conversation flow versus Rekognition media flow.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Lex and Rekognition serve different roles: Lex supports conversational interfaces, while Rekognition analyzes supported image and video inputs. They may be combined in a larger workflow when the use case requires both capabilities.

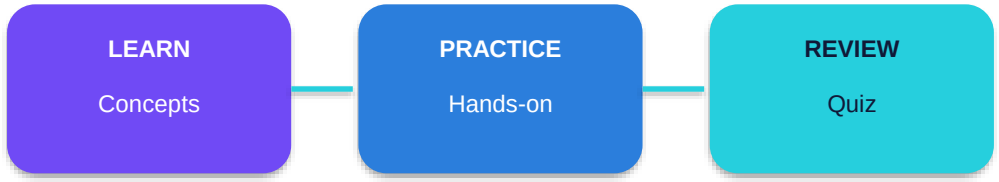
Mini Project – Conversational Media Assistant

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **User asks a question**
- Lex identifies intent
- Application receives media input
- Rekognition analyzes media
- Application returns a response

VISUAL CONCEPT | Chat interface connected to a media-analysis API.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A conceptual mini project can connect a conversational interface with a media-analysis workflow. The integration requires application logic, suitable permissions, error handling, and privacy review; it is not a built-in one-click integration.

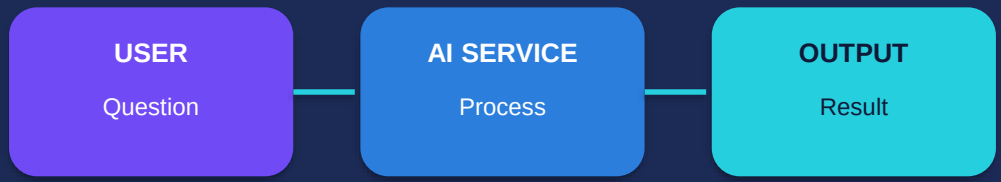
Conclusion – Building and Testing AI Experiences

Conversational AI, bot design, image and video analysis, and hands-on workflows

MAIN POINTS

- **Design clear intents**
- Collect only needed details
- Test varied conversations
- Validate media-analysis results
- Review security and privacy

VISUAL CONCEPT | Conversational AI and media-analysis learning roadmap.



INPUT → PROCESSING → USEFUL RESULT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

This presentation covered Lex dialog design and hands-on testing, Rekognition analysis workflows, and considerations for integrating AI capabilities into applications. Strong testing and responsible handling of data are essential.